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Artificial Intelligence Can Drive Sleep Medicine

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INTRODUCTION

The involvement of artificial intelligence (AI) in sleep medicine is reshaping our understanding of sleep disorders and their impacts on health outcomes (Fig. 1), as confirmed by a position statement[1] and a following commentary[2] from the American Academy of Sleep Medicine (AASM). Sleep medicine generates extensive physiologic data, serving as a rich resource for AI applications. However, despite significant advancements in AI, its impact on sleep medicine remains a buzzword for many clinicians. This article explores how AI can enhance the detection, analysis, and understanding of sleep patterns, emphasizing its potential to transform clinical practice. The article also delves into novel patterns in sleep, moving beyond conventional categorizations to explore concepts such as sleep depth and brain age. The development of large language models (LLMs) adds another layer of potential, enabling the extraction of insights from clinical narratives, and much beyond. AI also plays a critical role in elucidating the relationships between sleep disorders and health outcomes. By integrating multiorgan sleep signals, AI can enhance the accuracy of disease classification and offer prognostic insights, illuminating the complex interplay between sleep and overall health. We will explore these transformative applications of AI in sleep medicine, highlighting their potential to redefine diagnostics, enhance patient management, and ultimately drive sleep medicine.

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ACCESSING ARTIFICIAL INTELLIGENCE TOOLS FOR CLINICIANS

Before we introduce AI that can drive sleep medicine, we first introduce the often ignored and critical question: How do clinicians run the AI tools on their data? Despite the emerging AI tools, few reach the most beneficial group, clinicians. The reasons include the publication-centric view of the AI developers, the readership gap between AI and sleep medicine, and the lack of common knowledge and techniques between AI and sleep medicine. Researchers and policymakers have realized the gaps, and more policies are being made to bridge the gap. For example, the National Institute of Health (NIH) in the United States started to implement the Data Management and Sharing Policy in January 2023, which requires NIH-funded researchers to have a plan about the accessibility of the tool and dataset.

An increasingly popular approach for clinicians to access AI is through online Web sites, where AI researchers create a Web site, and clinicians can analyze their data by interacting with the Web site. For example, Moonlight[3] is an interactive viewer for sleep polysomnographic signals and annotations (https://remnrem.net/app/moonlight). Moonlight provides basic analysis abilities including automated sleep staging, visualizing spectrogram, comparing spectral band powers to the age-sex norm, and computing sleep electroencephalogram (EEG)-based brain age. Another example is U-Sleep,[4] which provides a Web site (https://sleep.ai.ku.dk/home) to perform nontraditional AI-based high-frequency sleep staging. The output, termed high-frequency hypnodensity, quantifies the sleep dynamics at a time resolution finer than 30 seconds, an epoch size that originated from the historical era of paper-based analog polysomnography. However, by uploading patient data, this approach could encounter data privacy and safety concerns. The concerns could be mitigated by de-identifying the patient data, including removing Health Insurance Portability and Accountability Act (HIPAA) information in the file name, meta-data, and annotations, and shifting sleep visit dates. The concerns could also be mitigated if the Web site supports ephemeral storage, that is, the uploaded data are only temporarily stored and are deleted automatically once the analysis or processing is finished. The clinician should also obtain approval from the regulatory committee in the health care institute for online AI data analysis.

Another increasingly popular aspect is smart devices, including wearables, nearables, and embeddables. Many people who visit the sleep laboratory have started to use AI-enabled wearable devices. The sleep clinician may indirectly access AI through their patient's use of a wearable device that reports sleep quality. More research is needed to show how clinicians can best use this information to aid clinical decision-making.[5] AASM has released a position statement regarding consumer smart devices for sleep.[6]

Another important aspect is AI education during sleep medicine residency training, which could be the most impactful approach to increase sleep clinicians' ability to access and use AI tools in the long term. It is an important venue for AI researchers to attend, lecture, and demonstrate basic AI concepts and how to use them, hence filling the gap at its root.

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A BRIEF INTRODUCTION TO ARTIFICIAL INTELLIGENCE

AI is a broad concept. Here, we focus on its machine learning (ML) and deep learning (DL) domains. These domains can be categorized into supervised learning, unsupervised learning, and reinforcement learning. Reinforcement learning requires an action, such as a robot moving in a maze, which is less relevant in this context. We focus on supervised learning and unsupervised learning.

Supervised learning is used when each data example is paired with a target label. The goal is to make the model output the correct answer for new, unseen data. Supervised learning includes classification and regression. For example, automated sleep staging is a classic supervised learning problem,[7] where each 30-second epoch is labeled as one of the 5 sleep stages. The same supervised learning paradigm can be applied to sleep staging with other signal modalities such as electrocardiogram (ECG) and respiration[8] and photoplethysmogram (PPG).[9] Many pattern recognition problems in sleep belong to supervised learning when the target labels are provided by humans, such as automatic sleep apnea detection and arousal detection.[10]

Some key techniques in supervised learning include linear regression, which is a linear regression model that predicts a continuous target, such as predicting apnea-hypopnea index (AHI) from ECG[11] or predicting brain age from EEG[12]; logistic regression, which is a linear classification model that assigns probability to discrete, unordered categories, such as classifying obstructive/central/mixed apnea for each apnea[13]; random forest and support vector machine (SVM), which is a nonlinear regression or classification model that allows to model complicated relationships. A special category of supervised learning relevant in sleep medicine is survival analysis, if the target label is time to an event, where the event can be death or a future diagnosis of disease.[14] The traditional Cox proportional hazard model can be technically counted as AI. More AI-centric models include deep convolutional survival analysis,[15] etc.

Unsupervised learning is used when there is no target label. Here, the task is to learn the representation and relationships within the data, such as hidden structures or clusters. A clustering example in sleep research is EEG microstates associated with dreaming.[16] Some key techniques in unsupervised learning include hierarchical clustering, where datapoints are repeatedly combined into larger and larger clusters based on similarity, which can be represented as a dendrogram; dimension reduction techniques such as principal component analysis, t-distributed stochastic neighbor embedding, and uniform manifold approximation and projection. Dimension reduction techniques have been used to visualize the landscape of sleep data.[17,18]

An important direction in unsupervised learning is self-supervised learning, which is widely used in the current era of large models. One popular example of self-supervised learning is contrastive learning,[19] where the model learns a representation invariant to transformations, such as changing the montage or applying filters. In the context of sleep signals, one can add noise to the signal and train the model to recover the original signal.[20] The model will learn the intrinsic structure of the data.

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Under unsupervised learning, generative AI (GenAI) has become more and more popular. GenAI is a particular class of DL algorithms geared toward generating new data that is synonymous with the data it has been trained upon.[21] GenAI algorithms typically consist of 2 neural networks: a generator and a discriminator. The overall goal of the GenAI algorithm is to learn the dynamics of the underlying data such that when a datapoint is generated by the generator, the discriminator cannot distinguish the generated datapoint as real or synthetic. As such, the GenAI models are powerful in that they can "generate" synthetic data that are as real as possible within the confines of the underlying dataset.

LLMs are a specific application of GenAI models. LLMs utilize the transformer architecture which has proved useful for numerous applications.[22] The seminal paper by Vaswani highlighted the transformer architecture for training LLMs that replaced existing neural network architectures.[22] Transformer architectures have recently been utilized for time-series data. At its core, the transformer architecture considers the context of surrounding elements of a given element. As an example, an LLM that is trained to learn language dynamics can interpret the sentence "patient denies having chest symptoms" as negative for chest symptoms as opposed to traditional natural language processing (NLP) models that would have attributed the sentence as positive for chest symptoms, in other words, the "denies" is not considered in traditional NLP whereas with LLMs the context surrounding words is equally important. In terms of sleep signal processing, the contextaware capacity of the transformer makes it suitable and achieves good performance when detecting patterns from EEG[23] and ECG,[24] as well as classifying sleep disorders.[25,26] We also expect to see large-scale applications of LLM in sleep research and medicine soon.

SLEEP DATA THAT CAN FUEL ARTIFICIAL INTELLIGENCE

Sleep medicine is notorious for generating large troves of data ripe for use in AI models. The saying "the bigger the better" is perhaps best fitting for AI models, and the field of sleep medicine is best equipped to fuel AI. Historically, nocturnal polysomnography (PSG) is the most important data source when analyzing an individual's sleep patterns. A typical full in-laboratory PSG contains EEG, ECG, electrooculogram (EOG), electromyography (chin and limb), respiration (flow, effort, snoring), and position, lasting 8 hours long and even with a modest sampling rate of 100 Hz (ie, a datapoint is acquired every 1/100th of a second). The total number of samples for PSG fed to an AI model is at the scale of millions. Home sleep apnea test (HSAT) is another form of sleep testing with fewer signal modalities. Given the ability of AI to do sleep staging and other tasks from nontraditional signal modalities, as well as the ease of performing HSAT and the mitigation of laboratory night effects, HSAT is another data source for AI applications.

In addition to PSG and home sleep tests, smart devices are also suitable for use in AI, which is increasingly popular. These smart devices typically focus on one modality (eg, photoplethysmography, accelerometer, etc).[6] Even though these wearables may not have all the data that a routine sleep study can acquire, these are still relatively high-resolution time-series data that can be used for AI.

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Another important data source for AI is the sleep reports and clinical notes written in natural language. Particularly, with the advent of LLMs, medical chart review and synthesis and reasoning of patient symptoms will be feasible using AI.

However, it is worth noting that most AI models utilize only a single or at most a handful of channels. It is still rare to see using AI to estimate the dynamics between the different signal modalities. For example, while researchers have studied the coupling between breathing and spindles,[27] AI models do not necessarily utilize that knowledge. Further, unknown physiologic concepts such as the time dynamics between periodic limb movements and perhaps arousals and/or breathing are also something that AI models can uncover.

ARTIFICIAL INTELLIGENCE-BASED STAGING OF SLEEP

Sleep staging is a classic task in sleep medicine, which has been well-studied in the AI literature. Some AI models are developed using target labels based on the AASM guideline, while some are based on the older Rechtschaffen & Kales (R&K) guideline.

Before we dive into the sleep staging models, the inter-rater agreement (IRA) among human scorers sets the human performance level which should be compared with AI performance. A meta-analysis of 11 international studies[28] showed an overall 5-stage Cohen's kappa at 0.76 (95% confidence interval, 0.71–0.81). The Cohen's kappa for W, R, N1, N2, and N3 are 0.70, 0.69, 0.24, 0.57, and 0.57, respectively. In Europe, the overall IRA among 8 centers was also 0.76 measured by Cohen's kappa.[29]

The state-of-the-art PSG-based AI sleep staging exceeds human performance. For example, L-SeqSleepNet, SleepTransformer, and XSleep-Net achieved overall Cohen's kappa at 0.83 on the large-scale Sleep Heart Health Study.[30,31] The performance of AI sleep staging depends on 2 factors, the training sample size and the model complexity. In the era of big data, the training sample size could be large enough, such as using datasets from the National Sleep Research Resource (NSRR)[32] and the Human Sleep Project under the Brain Data Science Platform (BDSP).[33] Therefore, we often see that state-of-the-art performance comes from DL, where the model complexity is higher compared with ML. Here, the difference between ML and DL is whether they use human-defined sleep signal patterns, that is, "features" in AI terminology, or they use raw sleep signals as the input without any feature engineering. ML requires human-defined features, including band power, band power ratios, spindle density, signal complexity measured by entropy, rapid eye movement, and EMG signal amplitude. On the other hand, DL uses raw signal as the input and automatically learns the features from the raw signal. Therefore, DL typically requires a larger sample size than ML. Because of the layered structure of DL models, the last hidden layer represents the learned features. Although DL is often thought of as a black box, it is possible to interpret the learned features,[34] including local approximation using linear models, occlusion, gradient visualization, activation maximization, and attention mechanisms.

However, the power of AI is not limited to using canonical signals for sleep staging. In addition, AI enables sleep staging from nontraditional signal modalities. For example, the

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state-of-the-art sleep staging performance using non-EEG signals includes overall Cohen's kappa at 0.75 using photoplethysmography (PPG),[35] and overall Cohen's kappa at 0.585 using ECG and respiratory effort belts.[8]

AI also enables finer-grained characterization of sleep stages. First, hypnodensity graph[36] is the probability version of hypnogram, where the *x*-axis is time, and the *y*-axis represents the probability of 5 sleep stages from an AI model, arranged vertically so that they sum to 100%. The probability outputs from the AI model are (or should be) calibrated so that they match the actual frequency of sleep stages with similar signal patterns as observed in the training dataset. Another novel information based on hypnodensity is "confidence," measured by the Shannon entropy of the stage probabilities, where the confidence is high when the probability is dominant for one stage, and low when the probability is close to 20% for all 5 sleep stages, such as during stage transition. The hypnodensity patterns have been used to identify narcolepsy patients.[36]

Second, high-frequency sleep staging methods, such as U-Sleep,[4] offer a finer-grained temporal resolution, where the staging can be done up to every 1/128 second. U-Sleep is based on a DL model that does time-series segmentation, instead of epoch-by-epoch classification, therefore, breaks the constraint of fixed epochs and greatly enhances the flexibility of sleep staging. Third, the recent surge of "foundation model" (still in preprints) may promote more accurate sleep staging. The foundation model is an AI model (often DL) trained on large-scale data such that it can be applied across a wide range of capabilities, including famous examples like ChatGPT. For sleep medicine, a foundation model may have broad capabilities in sleep staging, arousal detection, spindle detection, apnea detection, oxygen desaturation detection, flow limitation detection, etc. The broad capabilities of the foundation model indicate that the model has learned the basic representation of the data, which then supports the next step—fine-tuning. One can make the foundation model more specific to one or a few tasks by fine-tuning the later layers of the model with a small amount of training data. This approach brings the best of 2 worlds: the high model complexity and capability, and the relatively low amount of training data.

PATTERN DETECTION IN SLEEP ELECTROENCEPHALOGRAM MICROSTRUCTURE

In sleep medicine, some sleep EEG microstructures, such as sleep spindles, K-complex, and arousal, are only used for sleep staging. Other microstructures, such as cyclic alternating pattern[37] and infraslow oscillation[38] are often studied in sleep research, not in clinical practice. The sleep EEG microstructures contain rich physiologic information which is usually ignored. The definition is often qualitative and visual for many sleep EEG microstructure patterns. AI is good at extracting these qualitative and visual patterns by learning from human annotations using supervised learning. On the other hand, due to the qualitative and visual nature of sleep EEG microstructure patterns, there is a nonperfect IRA between human annotators, like those in sleep spindles.[39] Therefore, when we use the detection results from any detector, there is intrinsic uncertainty in the detection, onset time, and duration.

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Spindle Detection

Sleep spindle is defined as a train of distinct waves with a frequency of 11 to 16 Hz (most commonly 12–14 Hz) with a duration greater than or equal to 0.5 s, usually maximal in amplitude using central derivations. Sleep spindle is a hallmark of N2 sleep and also occurs in N3. Sleep spindle has been associated with memory reconsolidation and synaptic plasticity.[40] Another microstructure that is often together detected with the sleep spindle is the slow oscillation at the frequency range of less than 1 Hz, which is also called slow wave activity (SWA).

The IRA among humans represents human performance for sleep spindle detection. Using crowdsourcing to enable large-scale annotation, Warby and colleagues[39] assessed the IRA to be 0.75 ± 0.06 as measured by F1 score based on 110 healthy subjects, where each epoch was viewed by an average of 5.3 experts. In another IRA study,[41] the IRA ranges from 0.65 to 0.76 as measured by F1 score based on 180 subjects in the Montreal Archive of Sleep Studies.[42]

In terms of methodology, the traditional methods are based on signal processing, including short-time Fourier transform and wavelet transform, where the latter has higher temporal precision. Hilbert transform can also be used to get the instantaneous phase and amplitude of individual sleep spindles. The common software packages to detect sleep spindles include LUNA[3] and YASA.[43]

AI has demonstrated performance higher than human IRA. For example, the slim U-Net trained on massive online data annotation (SUMO) detector[44] obtained an F1 score of 0.82 on the same dataset as in the IRA study earlier. The SUMO detector has a similar U-shaped architecture to the U-Sleep model.[4] Another approach, utilizing the famous image segmentation model You Only Look Once (YOLO),[45] achieved 84% area under the precision-recall curve. It is also possible to detect sleep spindle in real-time,[46] achieving a high F1 score of 0.81, sensitivity of 83%, and precision of 78%, enabling feedback systems such as neuromodulation. AI-based sleep spindle detection also helps pediatric sleep medicine,[47] where the spindle morphology is not as well established as in adults.

In summary, sleep spindle detection has evolved from manual visual scoring to AI-driven methods that ensure high accuracy, repeatability, objectiveness, and scalability. The continuing improvement of sleep spindle detection with AI can improve our understanding of sleep and its impact on health.

Arousal Detection

Arousal is defined as an abrupt shift of EEG frequency, including alpha, theta, and/or frequencies greater than 16 Hz (but not spindles) that lasts at least 3 seconds, with at least 10 seconds of stable sleep preceding the change, during sleep stages N1, N2, N3, or R. Arousals reflect sleep fragmentation, depth, stability, quality, and underlying sleep disorders such as obstructive sleep apnea, insomnia, and restless legs syndrome. The development of AI-based arousal detection has been promoted by the availability of public large benchmark datasets, such as the PhysioNet 2018 “You Snooze, You Win” Challenge[48] and NSRR.[32]

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The winners of the PhysioNet 2018 challenge are Howe-Patterson and colleagues[49] and Li and colleagues.[50]

The IRA among humans represents human performance for arousal detection. In a study,[51] 10 experts from 7 centers scored 50 sleep recordings based on the AASM guideline. When compared second-by-second, the agreement among all scorers measured by Fleiss's kappa is fair (kappa = 0.40), while the pairwise agreements measured by Cohen's kappa vary from poor to substantial (kappa = 0.07–0.68). The agreement is similar in non-rapid eye movement (NREM) and rapid eye movement (REM). The similarity of the arousal index (ArI, number of arousals per hour) measured by the intraclass correlation coefficient is 0.41, also in the fair category. In summary, the overall agreement is low, which gives opportunities for AI to detect arousal objectively and consistently.

Many traditional ML models have been used to detect arousals,[52] including tree-based methods, SVM, K-nearest neighbors, and multilayer perceptrons. As ML models extract human-engineered patterns (features) from the signals, feature engineering and selection have been the crucial steps, which require significant domain expertise to develop a robust feature set.

With the availability of large benchmark datasets, increasingly DL models are used and achieve higher performance than ML models and are used by the PhysioNet 2018 challenge winners.[49,50] The models include convolutional neural networks, recurrent neural networks, U-net for segmentation, and transformers. DL models are end-to-end, where the input is the raw PSG signals, and the output is arousal detections. DL models can identify the subtle changes in EEG that characterize an arousal without explicit human feature engineering.

There are a few future directions for arousal detection in sleep medicine. For example, detecting arousals using non-EEG signals, such as in a HSAT setting. Josson and colleagues presented arousal detection from respiratory inductance plethysmography,[53] which records changes in the volume of the abdomen and chest. They showed that the AI model achieves a higher agreement than manual arousal scoring. Another direction is arousal in wakefulness after lights out, which is not counted in the current AASM guideline. The human IRA is the lowest in the wake stage (Fleiss' kappa = 0.25).[51] According to Wilson and colleagues, the exclusion of arousals in wake epochs leads to an underestimation of the arousal index.[54] The biological and clinical significance of arousal in wakefulness after lights out is understudied and requires more attention. Another future direction is real-time arousal detection and feedback systems, which could enhance patient monitoring and therapeutic interventions.

Respiratory Pattern Detection

Sleep apnea is one of the most common sleep disorders affecting over one billion people world-wide.[55] AI has been used to detect individual sleep apnea events, and severity metrics that can predict outcomes better than standard statistical models.[56–59] Recently, data from our group and others have shown that when the constituents of the AHI, a metric that is the clinical standard in diagnosing sleep apnea, are broken down and combined using AI, the resulting predictive models outperform standard measures and models. These models not

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only predict risk profiles but can also generate survival profiles that can highlight time points that are critical for intervention.

One of the drawbacks of standard regression or statistical models is that they do not account for the dynamics between the different covariates as well as the heterogeneity of the underlying datasets. AI models can overcome these limitations. In addition to the advanced AI models discussed earlier, much of the AI-based work in sleep apnea has been in detecting respiratory events (ie, apneas and hypopneas) and/or estimating the AHI. As such, the use of AI so far has been limited to time-series-based models that can detect specific patterns of interest. In the future, however, we can expect to see several studies that either utilize multimodal data (eg, electronic health record [EHR] data and time-series-based data) or multiple models that combine predictions across different outcomes given the technological advances in this field.

NOVEL PATTERNS IN SLEEP

Sleep Depth

The discrete nature of sleep stages is a result of human simplification. In NREM, the sleep stages form a continuum. Especially for N2 and N3, the cutoff of 20% SWA to define N3 and using 75 μV frontal peak-to-peak amplitude to define SWA are reasonable but still a choice by convenience. A continuous quantification of sleep depth can be implemented by AI.

A widely used sleep depth metric is the odds product ratio (ORP),[60] which is based on the distribution of band powers in EEG. The ORP ranges from 0 to 2.5, where 0 is the deepest sleep depth and 2.5 is fully awake. The ORP can be computed on the online Cerebra MY ORP platform, which also includes automated detection of many events. ORP has been found to change after continuous positive airway pressure (CPAP) treatment.[61] Other applications of sleep depth include measuring postarousal dynamics,[62] REM sleep depth,[63] and quantifying alpha intrusion.[64]

Although ORP is a simple distribution-based technique, it is possible to create ML or DL models with trainable parameters for sleep depth. By treating sleep stages as ordinal, that is, W<N1<N2<N3 ordered by sleep depth, one can train ML or DL models that learn the correct order.

Time-Frequency-Peaks as a Generalization of Sleep Spindles

The AASM definition of spindle frequency between 11 and 16 Hz may not hold for every adult. These transient oscillations can also occur in other frequency bands, such as alpha spindle which is associated with drowsiness.[65] Stokes and colleagues proposed a generalized concept: time-frequency-peaks (TF-peaks),[66] which defines transient oscillatory events in any frequency. While the biological and clinical significance of TF-peaks still needs to be clarified, it provides a novel foundation for characterizing a broad class of events with continuously varying spatial, temporal, and phase-coupling dimensions. The TF-peaks have been detected using traditional wavelet and multi-taper spectral estimation methods. Similar to SUMO for spindle detection,[44] they may also be detected using DL or ML methods.

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Large Language Model

The impressive capability of LLMs has brought a new surge of AI applications in many fields including sleep medicine. There are at least 4 directions of LLM in sleep medicine that were previously impossible. First, LLM extracts information from sleep reports, clinical notes, and neuroimaging reports. For example, various sleep parameters, current diagnosis, medical history, and family history can be extracted into tabular form from clinical reports written in natural language. Compared with traditional techniques such as n-gram and regular expression, LLM naturally handles negations, questions, and uncertainties in clinical reports. Second, LLM enables a natural language interface for the analysis and results of sleep data. The clinician may ask the LLM: "What is this patient's CPAP success rate?" Then, LLM will call the appropriate functions to calculate the success rate. The LLM-based language-to-code conversion has been proposed in various studies.[67] Recent studies have also shown LLM's capability to improve the function to achieve better performance.[68] LLM produces personalized insights and recommendations from sleep signal patterns and medical history. LLM integrates the multiple modalities in sleep medicine, and reasons the possible cause of observed abnormality,[69] therefore, reducing the time and effort for manual analysis. Last but not least, LLM holds the potential to improve access to high-quality sleep medicine health care in less developed areas.

ARTIFICIAL INTELLIGENCE-DRIVEN RELATIONSHIPS BETWEEN SLEEP DISORDERS AND OUTCOMES

Sleep-Based Brain Age Index

A key concept in sleep research is using sleep as a window into brain health and general health. For example, the sleep EEG undergoes profound changes with age, which provides a way to measure a "brain age" based on the sleep EEG pattern.[12] The brain age, or more broadly the biological age, matches our intuition of differential aging, where some people at 80 years old are still cognitively sharp, while others at 70 years old already experience many aging symptoms. One can compare brain age to chronologic age to define a brain age index (BAI). The sleep EEG-based BAI has shown to be longitudinal stable using 2 sleep recordings 5 years apart within the same person.[12] The night-to-night variability has been gauged and shown to reduce the variance in its estimation.[70] Elevated sleep EEG-based BAI is associated with dementia,[71] human immunodeficiency virus (HIV),[72] and mortality.[73]

Sleep-Based Classification of Existing Diseases and Prediction of Future Outcomes

Although the BAI has been associated with different outcomes, it is not specific. Just like other biological ages, such as bone age, an overly high BAI means something is wrong, but it does not tell the reason. There have been studies on the changes in sleep in different diseases. For example, many aspects of sleep are different in people with cardiovascular diseases such as atrial fibrillation,[74] spindle-slow oscillation coupling is deficient in diabetes,[75] spindle density is lower in schizophrenia,[76] lower sleep maintenance efficiency and longer wake after sleep onset are associated with lower cognition,[77] slow-wave sleep percentage negatively associated with incident dementia,[78] and REM slowing has been associated with different stages of dementia.[79] Many of these studies look at a

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single aspect of sleep. The AI approach holds the promise to use multiple sleep patterns, improving accuracy and enhancing interpretation. For example, Ye and colleagues used various sleep EEG patterns to classify different stages of dementia.[80]

Sleep can not only be used to classify existing diseases, sleep also predicts future events. For example, lower REM duration has been associated with mortality,[81] and lower heart rate variability during sleep has been associated with future Parkinson's disease.[82] The AI approach may extract subtle patterns even at subclinical stages. For example, Sun and colleagues estimated the incidence of 11 future outcomes, covering neurologic, cardiovascular, psychiatric, and mortality outcomes using AI and sleep EEG.[14]

SUMMARY AND FUTURE DIRECTION

The integration of AI into sleep medicine represents a paradigm shift to enhance the understanding and treatment of sleep disorders. The advancements covered in this article underscore AI's potential to not only surpass human performance in sleep analysis but also to facilitate the extraction of clinically relevant insights from complex physiologic data.

Future AI in sleep research should explore multimodal approaches that integrate various data types such as EEG, respiratory signals, electronic health records, and omics, to develop comprehensive models for diagnosing and managing sleep disorders. Future AI should also help clinicians to make decisions with data from wearable devices. This could lead to more effective monitoring and management of sleep disorders in daily life. More work should be done to create personalized treatment strategies based on individual sleep patterns, comorbidities, and lifestyle factors, using AI techniques such as reinforcement learning.

Importantly, as AI becomes more integrated into clinical practice, addressing ethical considerations and data privacy concerns is crucial. Future work should include developing frameworks that ensure patient data are handled securely while maintaining transparency in AI algorithms.

Finally, as mentioned in the beginning, AI education during sleep medicine residency training is critical and should be enhanced, which will ensure that the sleep field keeps pace with rapid advancements in AI.

DISCLOSURE

H. Sun: The author has nothing to disclose. A. Parekh: The author has nothing to disclose. R.J. Thomas: Licensed patent: ECG/PPG cardiopulmonary spectrogram. Develbiss-Drive: auto-CPAP algorithm patent and license. Patent submitted: Respiratory self-similarity for detection of high loop gain sleep apnea, enhanced expiratory rebreathing space (treatment of central/complex apnea). Unlicensed patent: positive airway pressure gas modulator for central/complex sleep apnea. General sleep medicine consulting: GLG Councils, Guidepoint Global. Dr. Sun and Dr. Thomas are supported by NIH R01HL161253. Dr. Parekh is supported by NIH R01HL171813, NIH R21HL165320, and NIH K25HL151912.

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KEY POINTS

- Artificial intelligence (AI) enhances the detection and analysis of known patterns in sleep medicine.
- AI discovers novel metrics and patterns in the sleep microstructure.
- AI provides prediction and stratification based on the sleep-health association.

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CLINICS CARE POINTS

- AI detection of patterns in sleep medicine has achieved similar or higher performance than human experts, such as sleep staging, apnea detection, arousal detection, and spindle detection.
- Although guidelines to standardize AI use in sleep medicine are yet to come, the clinicians can get more quantitative and deeper understanding of novel patterns, such as sleep depth, brain age, and adverse outcome risk, from the sleep signals using AI tools.

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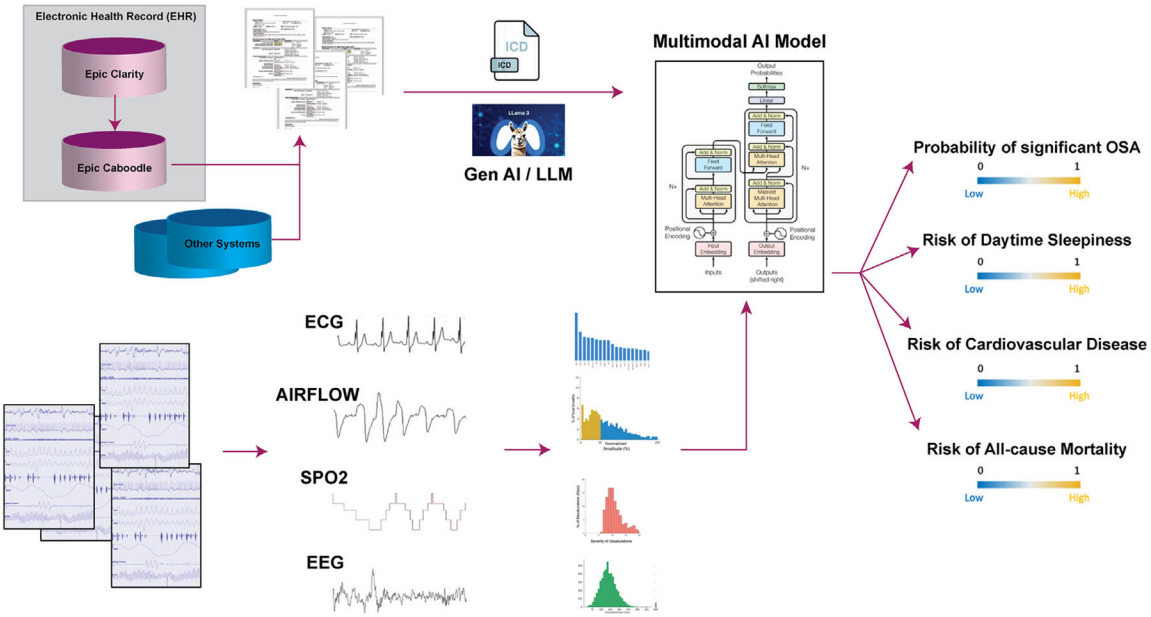

Fig. 1.
Overview of AI in sleep medicine.